

Augmenting Human-AI Co-Writing with Interactive Visualization

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ABSTRACT

Writing is a fundamental human activity—but today we have the opportunity to leverage Natural Language Processing (NLP) methods to help in this endeavor. Recent tools go beyond mere grammatical error-checking and use Large Language Models (LLMs) to support human-AI co-writing. While existing tools are helpful, many challenges remain: (1) mitigating ownership tensions between humans and AI; (2) enabling human autonomy in the process; (3) creating mechanisms for writers to understand and explore the reasoning of AI; and (4) applying NLP to complex and abstract narrative components (e.g., characterization, events, dialogue). In this paper, we hypothesize that some of these challenges can be resolved by introducing an communication interface between writers and AI. Further, we propose *Interactive Visualization*, a prominent method for making sense of complex AI reasoning and revealing hidden patterns from text data, to be that interface. To demonstrate our proposal, we present two case studies where we combine NLP and interactive visualization to support creative writing. The first case study is on mitigating social biases in fiction writing and the second is on the design of dynamic characters and scenes. We conclude by outlining our future work and broader impact.

KEYWORDS

Creative Writing, Fiction Writing, NLP, Visualization

1 BACKGROUND AND MOTIVATION

Modern creative writers use a range of computational tools. The first kind is text editor-like writing software such as Microsoft Word or Google Docs. There are several professional and paid software available to writers. For example, Scrivener [17] allows writers to organize a story in sections, add synopsis and notes to each section, and easily merge or swap between sections. Granthika [9] is a similar sort of paid service where writers can outline a more detailed narrative world, including character descriptions, major events in a timeline, and causal constraints on the events. These tools primarily enhance the organizational capabilities of writers with limited feedback on the actual writing.

More recently, researchers have proposed several emerging writing support tools. A dominant trend is the use of LLMs. These tools can generate text based on a prompt, often helping writers explore alternate narrative worlds and creative angles [2, 3, 8, 16, 21, 26]. Another set of tools uses NLP to extract patterns from the text that are otherwise difficult to notice. These tools typically focus on providing feedback while writers revise their text. For example, Du et al. [5, 6] proposed a tool to help writers in their iterative revising process. Serman et al. [23] proposed an analytic model that allows writers to interact with the literary style of an article. DramatVis



Figure 1: Supporting human-AI co-writing with visualization.

Personae [10] helps writers mitigate nuanced social biases in their writing. Dang et al. [4] integrated automatic text summarization in a text editor to help writers revise analytical essays.

While existing tools are useful to writers, there exist several challenges for human-AI co-writing. For example, one critical challenge is enabling human autonomy and reducing ownership tensions in the process [27]. Another relevant problem is the lack of trust among writers and AI as LLMs and other NLP models are typically black boxes and are difficult to interpret and explore intuitively [21]. Other challenges include identifying plagiarism and bias in the AI-generated text and applying NLP to more abstract and complex narrative components such as characterization, narrative structure, and events [8].

In this work, we hypothesize that some of these challenges arise from a lack of communication interface between writers and AI. We further propose interactive visualization to be that interface (Figure 1). There are abundant works on using visualization for text analytics [14, 24], human-AI collaboration [1, 11], explainable AI (XAI) [12, 25], and digital humanities [15, 18, 20]. However, it is still unknown how to translate and integrate these techniques to human-AI co-writing. Our work aims to fill that void. In the next sections, we present two case studies, demonstrating how NLP and visualization can be combined to support creative writing.

2 CASE STUDY 1: MITIGATING SOCIAL BIASES IN CREATIVE WRITING

Implicit biases and stereotypes are often pervasive in different forms of creative writing such as novels, screenplays, and children’s books [7, 13]. We interviewed 9 writers to understand how they mitigate unwanted biases in their writing. Writers mentioned that they actively look for such biases in their stories. However, these biases are often implicit and unconscious and difficult to wheedle out even for the best and most self-reflective of authors. The process is even more challenging for long and complicated stories where many characters take intersectional identities.

Based on the findings of the interviews, we designed DramatVis Personae (DVP) [10], a web-based visual analytics system to help writers identify stereotypes in creative writing. DVP is designed to integrate smoothly with the writer’s own creative process, allowing

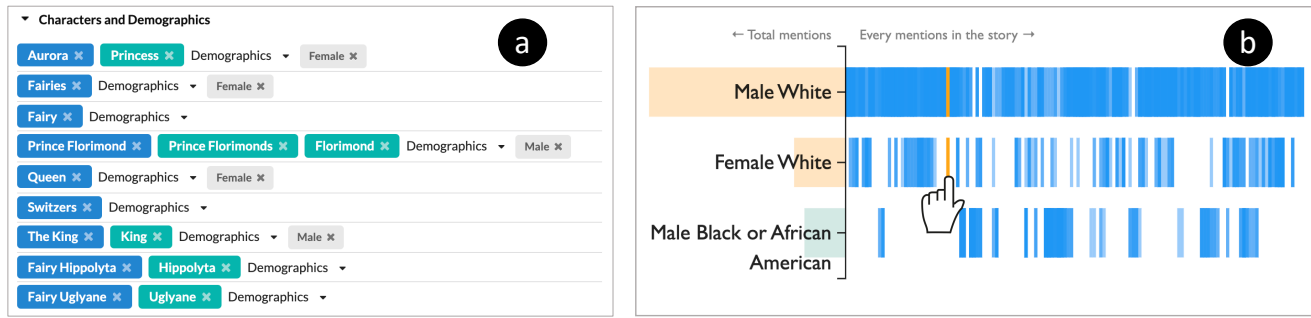


Figure 2: (a) Characters and Demographics panel of DramatVis Personae. A writer can assign social identities to their characters using this panel. In this instance, we are seeing the demographics panel for the children’s book, *Sleeping Beauty*. (b) An example timeline showing the presence of three intersectional identities (*Male White*, *Male Black or African American*, and *Female White*) in the Movie *The Amazing Spiderman 2 (2014)*. Male White characters are present throughout the storyline. Further, there appears to be a lack of interactions between Male African-American and Female White characters except for a few aberrant ones. However, both groups have interactions with Male White characters. The orange bars show one such interaction.

them to analyze existing literature for research, upload their written content as it becomes available, or even write in the tool itself, and then have its text analytics and visualizations updated in real-time. DVP automatically detects characters in a text and allows writers to assign social identities such as age, ethnicity, and gender to the characters (Figure 2a). The DVP dashboard uses this continually growing dataset to visualize the presence of characters and social identities over time (Figure 2b).

3 CASE STUDY 2: DESIGNING DYNAMIC CHARACTERS AND SCENES

Characters are often central to narrative fiction. Many writers base their stories on rich and complex characters that drive the plot, almost as if they have a life of their own. As a case in point, witness Walter White slowly transforming from milquetoast chemistry teacher to insidious drug kingpin in AMC’s *Breaking Bad* (2008). Designing such engaging and dynamic characters is challenging for writers as they need to believably show the evolution of the character throughout the story. While the process for developing characters may vary from writer to writer, we believe an analytic tool at the time of editing or revising a draft can help writers to identify discrepancies between intended and written text, decide what edits to make, and how to make those desired edits. For example, writers often try to portray an emotional journey for the characters (i.e., character arcs). Sentiment analysis can capture the character arcs and help writers revise their draft to achieve the intended arcs. Another common technique for creating dynamic scenes is to let characters with opposite emotions interact with each other (i.e., Bakhtin’s Polyphonic Theory). Sentiment and emotion analysis can again help writers design such dynamic scenes. We aim to integrate these models into a writing support tool using interactive visualization.

4 FUTURE RESEARCH DIRECTIONS

We envision several future directions for our research. We outline these directions below.

Empirical Studies. Our current works show promises for visualization as an interface between AI and humans. However, we have not properly evaluated the utility of visualization for mitigating the challenges listed in Section 1. One promising future direction is conducting empirical research to evaluate the potential of visualization in those scenarios. For example, we could integrate an XAI component like SHAP visualization [19] in a writing tool with a generative model. Writers can understand the reason behind the generated text from a prompt by investigating the visualization. We can study how such XAI component could help improving human autonomy and trust between writers and AI.

Designing Novel Visualization. Another promising approach is designing novel visualization to support complex creative writing tasks. Our first case study on mitigating social biases is one example of such research. Interactive visualization allowed us to integrate computational support for this sensitive and complex task. The tool empowered writers with access to advanced NLP methods without the need for algorithmic expertise. We encourage visualization researchers to invest efforts to design novel visual representations for other narrative tasks such as designing non-linear narrative structures and managing causal constraints between events, time, and settings.

Opportunities for NLP as an Analytic Partner. We believe our research has the potential to rekindle research on using NLP as an *analytic* assistant. This is in contrast with recent work on creative writing, which tends to focus on *generative* models [3, 16, 26]. However, we believe NLP as an assistant—when applied to reveal hidden patterns from stories rather than creating stories—still has powerful applications to offer. We hope our work will motivate future research to revisit how NLP can help writers analyze content. We believe that bridging the two directions—NLP both for story generation and for analytic assistance—is the right way to cement NLP’s importance in creative writing.

Scaling to Multiple Documents. The presented case studies support the analysis of a single document. We believe extending the

analysis to multiple documents will lead to several novel applications. First, writers could perform version comparisons so that they can easily see the impact of the edits. Support for such iterative revision has previously been shown to be useful [5]. Another potential direction is supporting the analysis of a text corpus. For example, we can answer questions such as *what is the representation of females in stories written during 1900?* Such analysis could be useful to literary scholars, who use language to understand society and the literature borne from it [22].

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